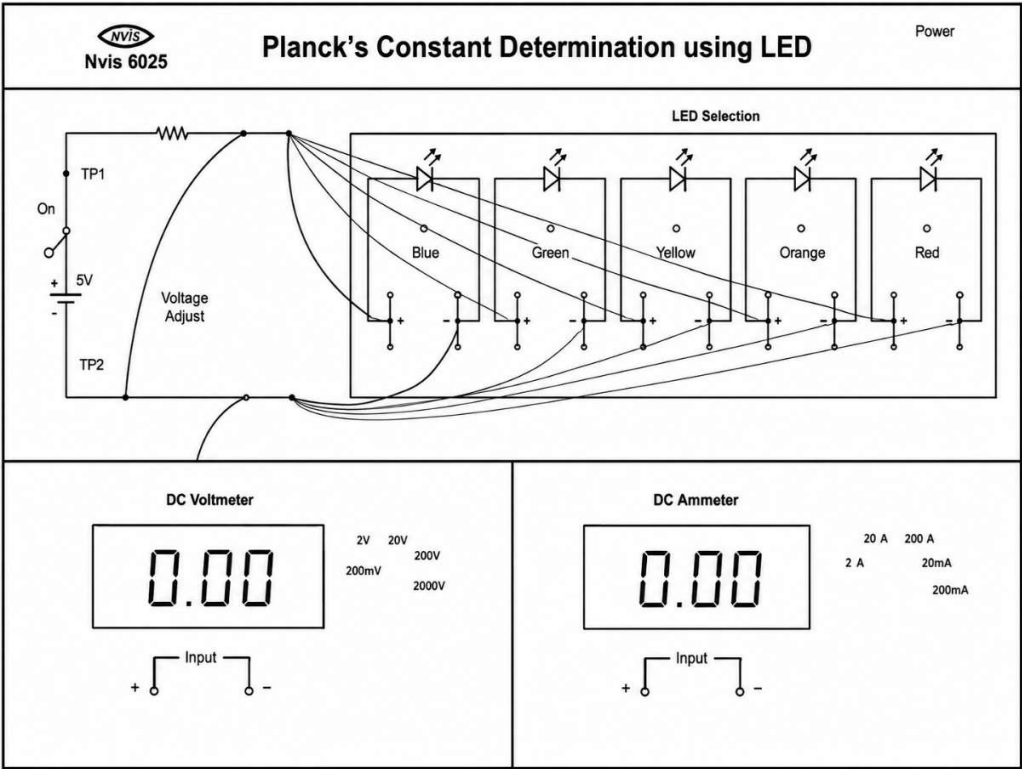


IILM UNIVERSITY, GREATER UNIVERSITY		LABORATORY MANUAL
PRACTICAL EXPERIMENT INSTRUCTION SHEET		
EXPERIMENT TITLE: Determination of Planck's constant using Light Emitting Diode (LED).		
EXPERIMENT NO. 03	DEPT. : Physics	ISSUE DATE: 12 August, 2026
LABORATORY: 26SOSP101 Semiconductor and Quantum Physics Lab	SEMESTER: I, II	TOTAL PAGE: 3
AIM: Determination of Planck's constant using Light Emitting Diode (LED). APPARATUS: Optical bench, meter stick, LEDs with 1000Ω resistor mounted (green, yellow, red), plastic holder, diffraction grating, power supply, multimeter + 1 lead, 1 alligator cable, 2 banana cables. CIRCUIT DIAGRAM: 		

THEORY AND FORMULA USED:

Planck's constant h is a fundamental constant of quantum physics. For example, it is used to describe the energy of a photon which is the smallest unit of energy of light of a given frequency. The photon energy is

$$E = hf = hc/\lambda$$

where h is Planck's constant, f and λ are the frequency and wavelength and c is the speed of light in vacuum. In this experiment, you will measure Planck's constant by measuring V_0 . The source of the photons is a light emitting diode (LED). A diode is an electronic device that allows current to preferentially flow in one direction. Current flows through the diode when the forward voltage across the diode exceeds a threshold value V_0 . At this voltage the energy of the emitted photon is $E = eV_0$ where e is the elementary electric charge.

$$h = \frac{eV_0\lambda}{c}$$

PROCEDURE:

1. Take the Planck's constant Determination Setup. Make the connections as shown in figure below.
2. Connect +ve terminal of DC power supply to +ve terminal of DC voltmeter and +ve terminal of any one LED.
3. Now connect -ve terminal of DC power supply to -ve terminal of DC voltmeter and -ve terminal of LED.
4. Set the range of DC voltmeter at 20 V. 5 Connect the mains cord and switch 'On' the power supply.
5. Connect the mains cord and switch 'On' the power supply.
6. Now vary the DC voltage slowly by variable resistance pot and see the LED connected in circuit.
7. When the LED just starts to emit light note the value of applied voltage by DC voltmeter.
8. Now switch 'Off' the DC power supply and break the LED connection.
9. Again, make same connection for another color of LED.
10. Now repeat the step 8 and 9.
11. Repeat above experiment for different colors of LEDs.
12. Now use the formula given below and put the value of all parameters used in formula and calculate the value of Planck's constant for different LEDs.

ANALYSIS:

Electronic Charge $e = 1.601 \times 10^{-19}$ coulomb

Velocity of Light $c = 3 \times 10^8$ m/s

LED	Threshold voltage V_0 (V)	Wavelength(nm)	$h = \frac{eV_0\lambda}{c}$ (js)
Blue			
Green			
Yellow			
Orange			
Red			

RESULT: The mean value of Planck's constant is.....

Percentage Error =

Precautions:

1. Make sure to keep a close watch on LEDs for activation voltage.
2. Take proper precautions while using electrical equipment.
3. Don't let too much current pass-through the LEDs otherwise it will be damaged.
4. Switch off the electrical equipment after work

Viva-Voce Question and Answers:

1. What is Planck's constant?
2. What is LED?
3. Why does LED emit the light in forward bias only?